

Project 6: Efficient Model Compression for Deploying Vision Models on Edge Devices

Abstract

Optimize a modern computer vision model for deployment on resource-constrained edge hardware (NVIDIA Jetson). The project focuses on navigating the accuracy-efficiency trade-off by implementing a model compression pipeline that includes techniques such as Knowledge Distillation and INT8 Quantization. The objective is to maximize inference throughput and minimize memory usage while maintaining competitive accuracy against the uncompressed baseline.

Expectations

- Baseline Deployment: Benchmark the original model's latency, FPS, and memory footprint on the target edge runtime (TensorRT/ONNX).
- Compression Pipeline: Implement a composite compression strategy combining Knowledge Distillation (Teacher-Student training) and Post-Training Quantization (PTQ/INT8).
- Pareto Analysis: Plot the Accuracy vs. Latency trade-off curve, comparing the Baseline, Distilled (FP32), and Quantized (INT8) variants.
- System Optimization: Ensure the deployment pipeline utilizes hardware acceleration (e.g., Tensor Cores on Jetson) for measurable speedups.

Hardware may use

- NVIDIA Jetson Orin NX (16GB)

Software may use

- PyTorch
- ONNX + TensorRT